

Review Article

Anticipated difficult airway during obstetric general anaesthesia: narrative literature review and management recommendations.

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Summary

We reviewed the literature on management of general and regional anaesthesia in pregnant women with anticipated airway difficulty. We identified 138 publications comprising 158 cases; these either described equipment or techniques for the provision of general anaesthesia, or the management of women with regional analgesia or anaesthesia, with the aim of avoiding general anaesthesia. Most of the former group described women requiring caesarean section alone, or in combination with other surgery, which was sometimes airway-related. Management techniques were largely similar to those in non-obstetric patients requiring surgery who have airway difficulties, although suggested differences related to physiological changes of pregnancy and avoidance of nasal intubation. In the reports discussing regional anaesthesia, consideration was often given to the possible requirement for urgent out-of-hours anaesthetic intervention, and the predicted difficulty of management of general anaesthesia should it be required. In a number of reported cases, multidisciplinary planning led to the conclusion that elective caesarean section should be performed in order to avoid emergency airway management. Based on this literature review, we advise antenatal planning that includes: assessment of the patient's clinical characteristics; consideration of the equipment and personnel available to provide safe airway management out-of-hours; and elective caesarean section should these be lacking. If general anaesthesia is required, a risk assessment must be made as to the probability of safe airway management after the induction of anaesthesia, and awake tracheal intubation should be used if this cannot be assured. Decision aids are provided to illustrate these points. Online appendices include a comprehensive compendium of case reports on the management of a number of rare syndromes and airway conditions.

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Introduction

Guidelines for the management of difficult and failed tracheal intubation in obstetrics were published jointly by the Obstetric Anaesthetists' Association (OAA) and the Difficult Airway Society (DAS) in 2015 [1]. These guidelines

provided a framework for handling unanticipated difficulty with airway management. However, other bodies such as the Canadian Airway Focus Group, have published recommendations for the anticipated difficult airway, albeit in all (non-obstetric and obstetric) patients [2].

Management of the pregnant women differs from other patients in several respects. Pregnancy implies that a potentially life-threatening event will occur within 9 months. In the UK, anaesthetists are involved in the care of around 60% of pregnant women [3], of whom approximately 10–15% of women will have an emergency caesarean section. However, not only is the time of delivery unpredictable for most women, but if severe fetal distress develops, the requirement to provide anaesthesia is arguably the most urgent of any surgical operation. Regional anaesthesia is used for most caesarean sections in the UK, but rapid sequence induction of general anaesthesia and tracheal intubation is the standard for the remainder. On the other hand, the treatment of primary airway pathology may occasionally be delayed or modified due to considerations of fetal well-being and the progress of pregnancy.

The aim of this review is to address the published evidence on management of the anticipated difficult airway in the pregnant woman, with special emphasis on antenatal multidisciplinary planning, awake tracheal intubation and other airway approaches that take into account the physiological and situational factors which are specific to the pregnant patient and the obstetric working environment.

Methods

We performed an electronic literature search in Medline, Embase, PubMed and the National Guidelines Clearing House for material published between 1970 and October 2018. The search terms used were: pregnancy, pregnancy complications, high risk, obstetrics, pregnant women, intubation, airway problem, awake intubation, awake fiberoptic intubation, awake laryngoscopy, awake tracheostomy, conscious intubation, awake LMA, conscious laryngoscopy, difficult intubation, anticipated difficult airway, anticipated difficult intubation known difficult airway/intubation, suspected or predicted difficult airway/intubation, rapid sequence induction, tracheal tube intubation, cricothyroidotomy, LMA, Proseal, Supreme, i-gel, videolaryngoscope, C-Mac, McGrath, Pentax, Kingvision, Glidescope, general anaesthesia, regional anaesthesia, continuous spinal anaesthesia, combined spinal epidural, airway assessment. There were no language restrictions.

The resulting list of 1976 publications was manually searched for relevant articles. We excluded case reports of women requiring surgery at before 12 weeks gestation.

Results

We have included 136 reports, both full publications and abstracts, covering 158 cases (see also Supporting

Information, Appendices S1–S6) [4–139]. Case reports largely fall into two groups: those describing equipment, or a technique, for provision of general anaesthesia; or those identifying a woman who had predicted airway difficulty, who was managed with regional analgesia or anaesthesia with the aim of avoiding of general anaesthesia. The findings are reported in these two groups.

Antenatal planning

A number of the cited publications were in the form of short reports, with limited information or describing the course of events but not the prior decision making. Many mentioned antenatal planning for delivery by a multidisciplinary team; when details were provided, the team sometimes included appropriate specialist surgeons and physicians (see also Supporting Information, Appendices S1–S6). Patient involvement in the decision making was mentioned in 38 reports.

Caesarean section was usually scheduled for obstetric indications, though some reports mentioned that it was chosen: to avoid labour or vaginal delivery [12, 14, 37, 41]; due to increased respiratory compromise in late pregnancy [20, 43–45, 63]; or in combination with definitive surgery for the airway lesion [17, 40]. In some patients, it was stated specifically that an elective caesarean section was performed because airway management was deemed too difficult should emergency caesarean section and general anaesthesia be required [33, 46, 50].

The location of delivery was sometimes dependent on other facilities; in some patients preparations for cardiopulmonary bypass or extracorporeal membrane oxygenation were made, requiring the use of a cardiothoracic operating theatre [49, 101, 104, 105, 134, 136].

In 40 out of 60 reports, where elective caesarean section was planned, this was carried out at the specified time. However, in 20 others, the woman underwent emergency procedures. The causes were obstetric (pre-eclampsia, preterm labour, concerns for the fetus, premature rupture of membranes) or worsening of the primary pathology, such as development of airway obstruction.

A plan for labour was described for 19 women. Advance planning included: the insertion of an epidural for analgesia, with the possibility of extending the level of the block should a caesarean be needed; or caesarean anaesthesia using awake flexible bronchoscopic intubation [88, 96], facemask anaesthesia [84] or rapid sequence induction with a rescue plan should initial intubation attempts fail [85]. Ten women required emergency surgery.

A thorough description of all potential delivery outcomes and anaesthetic options was provided by some

authors. Buist et al. considered the options for a woman with Klippel-Feil syndrome; the authors decided to allow spontaneous labour, and epidural analgesia was provided which aided the management of a forceps delivery [85]. Bamber and Evans described a woman with joint hypermobility syndrome, local anaesthetic insensitivity and a tongue tumour, for whom the authors chose elective caesarean section with awake flexible bronchoscopic intubation to avoid the possibility of emergency caesarean during labour or failed regional anaesthesia [46].

Of note, 22 women with fixed or chronic conditions that involved airway risks were only identified by the anaesthetic service at short notice before surgery, rendering advanced planning impossible [14, 20, 57, 65–67, 70, 71, 73, 76, 86, 91, 94, 98, 102, 104, 108, 109, 121, 132, 137, 139].

Airway assessment

An airway assessment was reported in 102 cases. A Mallampati grade [140] was recorded in 82 of these. Seven women had a modified Mallampati grade of 1; 17 a grade of 2; 33 were grade 3; and 30 were grade 4. All of the women with Mallampati grade 1 or 2 had other predictors of a difficult airway.

Other features / tests included: neck movement (mentioned 75 times); mouth opening (68); thyromental distance (36); and jaw protrusion/micrognathia (24). One report mentioned assessment of the neck in case front-of-neck access was required [50], and two reports mentioned obstructive sleep apnoea or snoring [15, 139]. No publication recorded the use of a combined score [141].

Previous difficulty with airway management was noted in 14 women [8, 9, 15, 28, 34, 35, 44, 46, 53, 54, 77, 84, 97, 121].

Specific pathology-related airway or respiratory investigations included two-dimensional airway reconstruction from CT images and intrathoracic anatomy [136]. In 15 women, awake laryngoscopy/nasendoscopy was carried out to assess laryngoscopy view before making a decision on anaesthetic technique [20, 23, 38, 39, 51, 53, 59, 60, 80, 96, 112, 130, 133, 137, 142], although most of these were before non-obstetric surgery.

Anaesthetic management

Awake tracheal intubation using the flexible bronchoscope

We identified 52 women having awake flexible bronchoscopic intubation, all but eight of whom were at caesarean section [7, 11, 24, 25, 29, 31, 38, 42]; five women also had thyroidectomy performed under the

same anaesthetic (see also Supporting Information, Appendix S1). Three women underwent flexible bronchoscopic intubation after an initial direct laryngoscopy showed a poor view [9, 20, 38]. In five women, bronchoscopic intubation followed failed regional anaesthesia [10, 14, 16, 27, 35].

One case involved two stages of surgery, whereby an epidural was used for caesarean section, and then awake fibreoptic bronchoscopic intubation was used for thyroidectomy. The femoral vessels were prepared for cardiopulmonary bypass, and topical local anaesthesia was applied to the patient's airway before the caesarean started [39].

The majority of women had congenital abnormalities, but other indications were: extreme obesity [11, 16, 21, 27, 48]; goitre [12, 17, 23, 40, 49]; mediastinal mass [13, 25]; HELLP syndrome [6, 9]; a bitten and swollen tongue following an eclamptic fit [4]; and limited mouth opening [5, 7, 16, 18, 19, 29, 30, 35, 48]. The use of sedation and topical anaesthesia techniques was highly variable. Oxygen was provided by facemask, nasal cannulae or by high-flow humidified nasal oxygenation at 50–60 l.min⁻¹ [49, 50].

There were three cases of complete failure, although another patient had successful oral intubation after failed nasal attempts [32]. Of the three cases, two patients' tracheas were intubated with a videolaryngoscope following induction of general anaesthesia [44, 45] and one died from hypoxia, no further details being reported. [26]. Cases of flexible bronchoscopic intubation after unanticipated failed tracheal intubation have been reviewed previously [143].

Awake tracheal intubation using other devices

Four reports described an initial awake direct laryngoscopy; in two patients, intubation was performed awake, whereas general anaesthesia was induced before intubation in the other two [51, 53]. A number of devices were used to secure the airway with the patient awake (see also Supporting Information, Appendix S2). Chronologically, these reflect a progression from the conventional Macintosh laryngoscope to newer devices, including videolaryngoscopes for laryngoscopy and supraglottic airway devices as a conduit for intubation. A common observation in all the above cases was that the patients were morbidly obese, and the anaesthetist expected difficulties in tracheal intubation and facemask ventilation. Videolaryngoscopy has also been used as a primary technique or after failure of other approaches [50, 61].

Full topical anaesthesia of the airway before awake laryngoscopy was explicitly or implicitly suggested in these

case reports. However, Kariya et al. reported intubation with lidocaine jelly applied to the laryngoscope blade as the sole topical anaesthesia in two patients [144]. The authors suggested that glottic views with the Pentax™ (PENTAX Europe GmbH, Hamburg, Germany) laryngoscope was less stimulating than with direct laryngoscopy; however, these women required conversion of regional to general anaesthesia due to haemorrhage during caesarean section, and hypovolaemia and hypotension might have affected the requirement for local anaesthesia.

Airway control after induction of general anaesthesia

Reports in which general anaesthesia was induced before securing the airway mainly feature the use of a videolaryngoscope or supraglottic airway; the latter usually being used to aid tracheal intubation (see also Supporting Information, Appendix S3). In some patients, this followed failed awake flexible bronchoscopic intubation [44, 45]. As mentioned above, the steps of laryngoscopy and intubation were sometimes separated out, with general anaesthesia induced only after a good view of the larynx was seen [51, 53].

Tracheostomy

Subglottic stenosis during pregnancy has been managed successfully with tracheal dilation or resection of the lesion, and hence the further obstetric course was unaffected [142, 145–148]. However, tracheostomy has been used to manage the airway during surgical resection of obstructive lesions (see also Supporting Information, Appendix S4). It has also been used pre-emptively to provide a secure airway during labour or delivery; for instance, it may be decided that airway stenosis is not compatible with the increased ventilatory demands of the developing pregnancy or labour, or even safe regional anaesthesia at caesarean section, as occurs in patients with spinal muscular atrophy [83] or subglottic stenosis [78, 82].

Pre-emptive tracheostomy during pregnancy was described in the management of a patient who had a history of failed intubation, and an unfavourable airway examination [77]. She presented at 32 weeks gestation with haemorrhage from a placenta praevia. After multi-disciplinary discussions, it was agreed that an elective tracheostomy using local anaesthesia would be the safest option as, in the event of significant placental haemorrhage, it would not be appropriate to take 5–15 min for an awake flexible bronchoscopic intubation before inducing general anaesthesia. In the event, after 18 days as an inpatient, she started having contractions and had a minor bleed; expedited caesarean section was performed using epidural anaesthesia.

Tracheostomy has also been used, following a difficult awake flexible bronchoscopic intubation, to provide a secure airway for further management, until resolution of supraglottic oedema in a patient with Crouzon's syndrome [27].

Clearly, the patient has to agree to having the procedure performed, and we found reports where women had refused consent for tracheostomy for management of airway surgery [142, 147] or caesarean section [115].

Regional analgesia during labour

Cases where women with predicted airway problems were managed using regional analgesia for labour are shown in the Supporting Information (Appendix S5). Successful surgical conversion for operative vaginal delivery [85, 87, 96] and caesarean section [86, 88] were noted. Unsurprisingly, patients requiring general anaesthesia were also described [27, 53, 65, 73]. Some patients presented with specific features that made regional anaesthesia more difficult, including short stature, spinal abnormalities, previous back surgery and obesity. Failure of epidural analgesia was attributed to the medical condition in a woman with ankylosing spondylitis [94]. At the first attempt, the epidural needle location was confirmed by obtaining cerebrospinal fluid through a long spinal needle passed through the epidural needle [149], but neither this, nor the placement of a second epidural catheter provided analgesia. Eventually, continuous spinal analgesia was provided by subarachnoid placement of an epidural catheter, which provided analgesia for the duration of the labour [94].

Regional anaesthesia for caesarean section and surgery

Women who had regional anaesthesia initiated for surgery are included in the Supporting Information (Appendix S6); all but one had a caesarean section, combined with another operation in several cases. As in the section 'Regional analgesia during labour' above, there were many patterns of failure to establish anaesthesia. General anaesthesia was used in urgent cases [67, 73] and for intra-operative bleeding after delivery of the fetus [107].

The indication for choosing a particular regional anaesthetic technique was rarely mentioned explicitly. Combined spinal-epidural anaesthesia was selected in some patients for its perceived reliability and the ability to 'top up' the epidural catheter [118, 125], or to allow small dose spinal initiation [45, 121, 123, 135]. A woman with Klippel-Feil syndrome was provided with combined spinal-epidural anaesthesia; in a preceding meeting of anaesthetic staff that considered the advantages and disadvantages of general anaesthesia and various types

of regional anaesthesia, no consensus was reached on the best option [125].

Inhaled helium was used to reduce the work of breathing [92, 99, 107], and high-flow nasal oxygenation at 40 l.min⁻¹ was provided during surgery to a woman who had subglottic stenosis [133].

Medical problems causing difficulty with airway management

The management of women with specific airway diagnoses or problems, including obesity, varied significantly. Appendix S7 shows published information classified in this format.

Discussion

Antenatal planning

A few case reports described early antenatal referral to the anaesthetic service, multidisciplinary discussions and meticulous planning that considered the interaction of the potential obstetric course of events and associated anaesthetic requirements [85, 103]. Patient involvement in the decision making, mentioned in some reports, is crucial in order for the woman to appreciate the risks, understand the decision making process and give their consent to the final plan.

Some cases might have featured good practice that was not mentioned in the publication; however, it is also apparent that patients may be referred for anaesthetic assessment very late in pregnancy, or even after a decision for emergency surgery has been made. Cases of women having caesarean section are over-represented in the literature; in this situation an anaesthetist will definitely be involved, whereas women undergoing labour with potential airway problems may not have had the same consideration, or may not be known to the anaesthetic service.

If an elective caesarean section is planned, a contingency must also be prepared in case emergency caesarean is required after early onset of labour or obstetric complications. Some units may have the facilities to perform (awake) flexible bronchoscopic intubation out-of-hours (this usually being seen as the safest method to manage the woman with extreme airway difficulty). However, the time required, even in expert hands, is likely to be significantly longer than for rapid sequence induction [150]. It is clear that there is not a 'recipe book' approach to more complex cases, as the patient, professionals, practice and facilities will all vary on an individual basis [46, 85, 125].

Women who require airway surgery in addition to caesarean section, or in whom extracorporeal oxygenation

methods are planned, may have to be delivered in a different unit such as a cardiothoracic centre [49, 101, 104, 105, 134, 136], or an operating theatre away from the delivery suite [40].

Airway assessment / definition

This review is largely based on individual case reports, in which the author defined the patient as having a potential difficult airway. Reporting bias is highly likely, although this cannot be further qualified. The Canadian Airway Focus Group provided definitions for a 'difficult airway' and its individual components, encompassing problems encountered with facemask ventilation, insertion of supraglottic airway devices, laryngoscopy, tracheal intubation and front-of-neck airway access [151]. However, prediction of a problem is an uncertain exercise, especially as the course of events will be determined by a complex combination of factors involving the patient, anaesthetist, equipment and environment. In a general population, prediction of difficult intubation and mask ventilation is very insensitive and poorly specific [152], and combined tests add little to single tests [141]. In the general obstetric population, the factors associated with difficult intubation appear to be the same as for non-pregnant patients, including high Mallampati score, short neck, receding mandible, protruding maxillary incisors [153] and increased neck circumference [154].

Tracheal intubation is a routine component of obstetric general anaesthesia in many countries. Hence, in the literature on anticipated airway difficulty, the antenatal assessment usually focuses purely on difficult laryngoscopy and tracheal intubation. Cases were identified mainly based on the presence of a syndrome or specific anatomical abnormality, or obesity. The Mallampati score was the most frequently used single test for difficult laryngoscopy. Interincisor distance and neck movement were reported less frequently. However, it is not possible to recommend a test or group of tests based on this literature review.

A minority of cases described conventional airway assessment in situations where the laryngoscopy might be straightforward but intubation difficult, such as subglottic obstruction including thyroid enlargement, or mediastinal lesions that might cause problems with positive pressure ventilation.

Potential difficulties with mask ventilation were explicitly considered in some case reports, usually leading to a decision for awake intubation. A number of studies have investigated the risk of difficult or failed mask ventilation in

general populations [155, 156], but not in the obstetric population. Several of the risk factors identified in those studies do not apply to obstetric practice, including male sex, age and presence of a beard. Those that are relevant include raised body mass index (BMI), neck circumference, Mallampati grade and reduced thyromental distance [1].

A consideration that did not feature in the literature was the possibility of the use of a supraglottic airway device in the event of difficult or failed intubation. Most reports date from after the introduction of the laryngeal mask, but the specifics of secondary (failed intubation) plans were rarely mentioned. Restricted mouth opening and the presence of supraglottic airway pathology are the most important barriers to the use of a supraglottic airway device as a conduit for intubation.

Anaesthetic management

Awake tracheal intubation

Awake intubation is advised when it is considered that oxygenation and manual ventilation may not be guaranteed after induction of anaesthesia in a woman who might be difficult to intubate. The reasons for this include: failure to achieve a tight seal between facemask and face; upper airway collapse as a result of general anaesthesia; and/or poor chest compliance. The majority of morbidly obese women in this literature review requiring general anaesthesia had their airway secured with an awake technique, either using a flexible bronchoscope or videolaryngoscope. Non-surgical techniques for awake intubation include direct laryngoscopy, videolaryngoscopy, flexible bronchoscopic intubation and via a supraglottic airway device. High-flow nasal oxygenation has been shown to improve oxygenation during flexible bronchoscopic intubation in non-obstetric cases [157]. A modified approach to establishing the airway is to perform initial awake laryngoscopy with topical anaesthesia, with the option of inducing of general anaesthesia if there is a satisfactory view of the larynx [51, 53].

Flexible bronchoscope

There are no definitive figures for the overall frequency of awake flexible bronchoscopic intubation in obstetric general anaesthesia in the UK. We have made an indirect estimate based on data from a 2014 survey of the number of failed tracheal intubations and awake flexible bronchoscopic intubations in the previous year in UK obstetric units; 55 and 24 cases respectively were reported [158]. Based on a rate of failed tracheal intubation of 2.6 per 1000 cases [143], this would give a rate for awake flexible

bronchoscopic intubation of 1.1 per 1000 for obstetric general anaesthesia in the UK. This is similar to the findings of McDonnell et al. in Australia [159], who recorded one awake flexible bronchoscopic intubation, two inhalational inductions and four failed intubations after rapid sequence induction among 1095 caesarean sections under general anaesthesia. A significantly higher awake flexible bronchoscopic intubation rate of 10.1 per 1000 was recorded by Rajagopalan et al. in the USA [48]. We can probably conclude that the use of awake flexible bronchoscopic intubation does not closely reflect the degree of airway difficulty, but rather the skills of the anaesthetists and availability of equipment. The route for awake flexible bronchoscopic intubation was not stated in a number of case reports; we can assume that oral intubation was performed in around half of the cases, or more.

Data spanning 35 years were provided by Glassenberg, who attempted to reduce the incidence of failed tracheal intubation within his obstetric unit with a liberal use of sedated flexible bronchoscopic intubation during the middle period of data collection. Between 1974 and 1985, awake flexible bronchoscopic intubation was not used at all, but was then performed in 14% of all caesarean sections under general anaesthesia between 1985 and 2004, and in less than 5% between 2005 and 2010. He concluded that an awake flexible bronchoscopic intubation rate of 15% would be required to halve the incidence of failed tracheal intubation, and suggested that a more pragmatic approach is a combination of videolaryngoscopy during rapid sequence induction of general anaesthesia, and laryngeal mask use after failed tracheal intubation [160].

Other devices

A recent review of non-obstetric practice suggests that awake videolaryngoscopy provided equal success rate and complications, but shorter intubation times, when compared with fiberoptic bronchoscopy [161].

Airway control after induction of general anaesthesia

Recent literature on high-flow humidified nasal oxygenation suggests that this technique maintains good oxygenation in the apnoeic non-pregnant patient for a prolonged period of time. The 2015 OAA/DAS obstetric intubation guidelines and 2015 DAS non-obstetric intubation guidelines advised that high-flow humidified nasal oxygen be considered as a standard technique for procedural oxygenation at induction of general anaesthesia [1, 162]. This would be even more applicable for women with a predicted difficult airway.

Recent work in pregnant women has suggested that pre-oxygenation with high-flow humidified nasal oxygen, assessed using end-tidal oxygen concentration, may not be as effective as the standard pre-oxygenation technique with a facemask, although these studies did not investigate the effectiveness of high-flow humidified nasal oxygen in extending safe apnoea time [163, 164].

In non-obstetric cases, videolaryngoscopy may well replace direct laryngoscopy as the first choice device in the patient with predicted difficult intubation [165, 166]. It may also replace flexible bronchoscopic intubation as a lower skill method for managing most patients who do not have airway lesions or pathology [167]. The use of a combination of videolaryngoscope and flexible bronchoscope together has also been described [168]. These considerations should also apply in the obstetric patient.

If there is a strong likelihood that rescue front-of-neck airway access might be required, consideration should be given to performing an ultrasound scan of the neck, and marking the cricothyroid membrane before induction.

Tracheostomy

If the patient has a subglottic stenosis, the anatomy of the supraglottic airway is usually normal, and therefore tracheal intubation may be performed as usual; however, passage of even a small tracheal tube may be difficult. A fixed obstruction presents problems, especially when minute ventilation is increased, such as during labour. Elective caesarean section may be required [130], though with appropriate safety considerations it may be acceptable to allow labour [96].

In the past, an ear, nose and throat surgeon was often considered the most appropriate person to perform an emergency tracheostomy. However, the decline in use of surgical tracheostomy, together with the increase in percutaneous tracheostomy techniques in ICU, means that the anaesthetist may be the most skilled person to carry out an emergency scalpel cricothyroidotomy [169].

Regional analgesia during labour

Unless contraindicated, regional analgesia was chosen to aid the management of labour when this was planned. The ability to extend an epidural blockade for operative delivery was usually mentioned, although a back up plan for managing general anaesthesia in the event of top up failure was sometimes stated [87].

There is suggestive evidence that the needle-through-needle combined spinal-epidural technique is more reliable than epidural alone [170], as long as the

spinal component is successful. A free flow of cerebrospinal fluid through the spinal needle suggests that the loss of resistance felt with the epidural needle denotes correct midline placement within the epidural space, and will make inadequate spread of local anaesthetic when the blockade is extended for surgery less likely.

A meta-analysis showed that the factors associated with failure to extend labour regional analgesia adequately for caesarean section included: an increased number of analgesic boluses during labour; increased urgency category for surgery; and a non-specialised (obstetric) anaesthetist providing care [171]. This analysis included studies where boluses were administered by a care provider for breakthrough pain. Patient-controlled epidural analgesia and programmed intermittent epidural bolus techniques have recently gained much popularity, but there is no information to relate the use of these techniques to success of an eventual surgical top up.

Continuous spinal anaesthesia has a number of theoretical advantages. However, lack of experience with its use during labour, lack of microspinal equipment availability in some countries, and the risk of post-dural puncture headache with macrospinal techniques, ensure that this is currently a technique of last resort.

The preceding discussion relates to women with normal anatomy, and individualised decisions have to be made in women who pose potential or actual problems with block placement.

Regional anaesthesia for caesarean section and non-airway surgery

A variety of methods were used in this situation. Catheter techniques that allow incremental doses and prolongation of anaesthesia are theoretically preferable. However, epidural anaesthesia uses high local anaesthetic doses, is slow and provides poorer sacral blockade. The combined spinal-epidural technique led to a number of failures in the intended technique. Two anatomical spaces have to be located, although this is assisted with the needle-through-needle combined spinal-epidural technique, as the epidural needle acts as a guide for the spinal. On the other hand, this technique means that the epidural catheter may not be tested until it is too late to correct its position [125].

Continuous spinal anaesthesia for surgery has more of an evidence base than for labour analgesia; but again, it is unfamiliar to many practitioners, and microspinal equipment is not available in some countries.

As with labour regional analgesia, the course of action may have to be altered if there are anatomical abnormalities, whether predicted or not. Some specific issues that may have to be considered include: the availability of an appropriately low interspace and the possibility of spinal cord or cauda equina tethering for spinal anaesthesia; previous back surgery; the absence of an epidural space; short stature; the risk of a rapid onset or high block; the risk of cerebrospinal fluid leak in patients with raised intracranial pressure; and high BMI.

Dresner et al. assessed the possibility of direct laryngoscopy using local anaesthetic, with the intention of performing awake flexible bronchoscopic intubation should it appear to be difficult, in order to avoid sudden requirement for airway control. In the event, laryngoscopy was straightforward and regional anaesthesia was performed [138]. Kavanagh et al. also provided topical local anaesthesia to the airway, but on reviewing their case afterwards, the authors concluded that this was mistaken; by the time that intubation would have been required, the local anaesthetic application would have had to be repeated [125].

Practice recommendations

How is the obstetric patient different from the non-obstetric patient? There are some differences in the anatomy and physiology of the airway, such as mucosal engorgement and friability, and susceptibility to hypoxaemia. The overriding difference is that the obstetric patient is much

more likely to present for major surgery that does not relate to the airway itself, and this may be unpredictable, time-pressured and/or associated with physiological upset from significant haemorrhage. In the ideal situation, anaesthesia will be provided by an anaesthetist with the relevant specific airway skills and the appropriate surgical teams. However, if labour is intended or allowed, there will always be the possibility of requiring an emergency caesarean section or other surgery, which may happen out-of-hours; a plan should therefore be made for anaesthesia to be administered at short notice by an anaesthetist with limited or no advanced airway skills.

Antenatal planning

The recent guideline from the UK National Institute for Health and Care Excellence (NICE), '*Intrapartum care for women with existing medical conditions*', does not cover women with potential difficult airway problems. However, the general recommendations are relevant, stating, "*A multidisciplinary team led by a named healthcare professional should involve a pregnant woman with a medical condition in preparing an individualised plan for intrapartum care*" [172].

A pregnant woman with a known or suspected difficult airway, or a history of airway-related anaesthetic problems, should be referred antenatally for anaesthetic assessment in order to fully investigate the extent of the problem (Fig. 1). Factors in the medical history that might alert to a potential airway problem include: head, neck or jaw surgery,

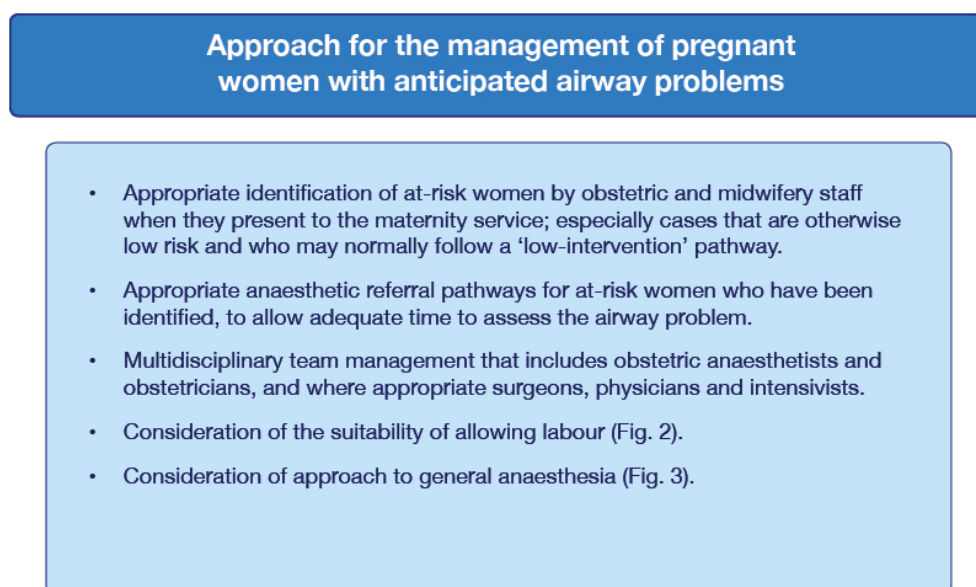


Figure 1 Suggested approach for the management of a pregnant woman with potential airway problems.

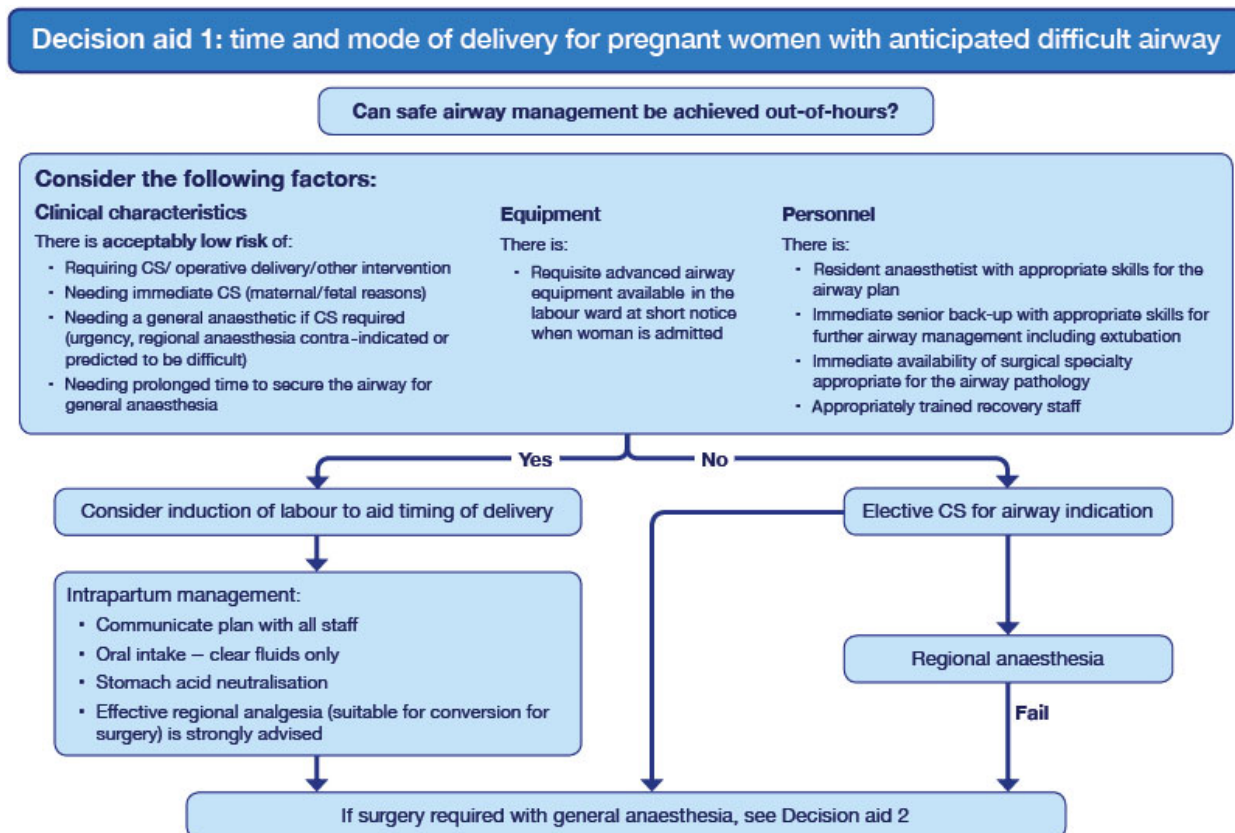


Figure 2 Decision aid 1: time and mode of delivery for a pregnant woman with anticipated difficult airway. CS, caesarean section.

tumours, pathology etc; morbid obesity; obstructive sleep apnoea; juvenile rheumatoid arthritis; mediastinal tumours; and some congenital syndromes [173]. Of note, a number of cases of upper airway obstruction were misdiagnosed and treated as asthma, in some cases for several years [13, 76, 100, 105, 112, 142].

Some women with extreme or specific airway difficulties may require transfer of care to a specialist centre for management of delivery. A multidisciplinary team meeting might include anaesthetists, obstetricians, midwives and other specialists members such as ENT, oncologists and cardiothoracic surgeons, and include the woman where appropriate. This should usually be carried out in the middle trimester, to allow for premature delivery or worsening of symptoms in later pregnancy. If a formal approach to assessment of risks seems merited, then decision tree analysis might be considered [46]. The approach should include primary and back up plans for mode of delivery and anaesthetic management. In some women, elective caesarean section may be recommended to ensure anaesthesia and airway management is provided

by experts and within normal working hours (Fig. 2). These plans should be readily available in the woman's maternity notes; in the UK, women carry their own records so that they are available should the woman present with a problem elsewhere than her booking hospital.

Labour

The plan for labour should take account of the airway skills and equipment available out-of-hours, rather than the highest level of anaesthetic expertise in the hospital. Figure 2 indicates the factors that should be considered. Labour management should aim to ensure that any surgical intervention is not required with category 1 urgency that does not allow appropriate time to establish regional or general anaesthesia safely. This may mean that operative delivery is performed earlier than would otherwise be the case. Airway swelling may worsen during labour, thus making tracheal intubation more difficult. Significant factors include pre-eclampsia and prolonged active pushing in the second stage.

Establishment of regional analgesia is thought to reduce the likelihood that general anaesthesia will be required for a

Decision aid 2: general anaesthetic approach for pregnant women with anticipated difficult airway

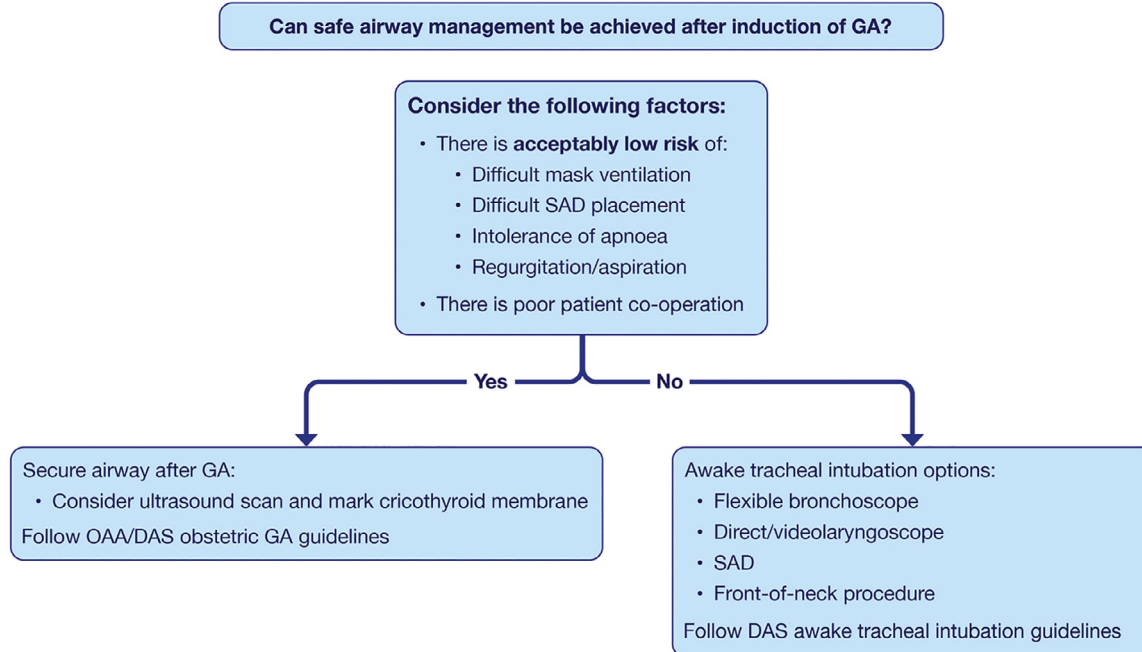


Figure 3 Decision aid 2: general anaesthetic approach for a pregnant woman with anticipated difficult airway. GA, general anaesthesia; SAD supraglottic airway device; DAS, Difficult Airway Society; OAA, obstetric anaesthetists' association.

subsequent surgical procedure. The use of ultrasound may assist in siting regional analgesia [174]. A technique that involves needle-through-needle location of cerebrospinal fluid (combined spinal-epidural or dural puncture epidural) may be more reliable than standard epidural analgesia. However, the most important aspect with any regional analgesic technique is to assess thoroughly (including dermatomal block) and optimise, in order to ensure the best chance of successful top up for surgery [175, 176].

Specific preparation for a general anaesthetic may include the use of topical anaesthesia to the airway, although this would be difficult to time in advance to ensure adequate effect. Identification and marking of the cricothyroid membrane is possible, although the accuracy of advance marking has not been assessed.

Caesarean section

Regional anaesthesia for caesarean section is usually reliably effective, especially in elective cases [177]. However, some women who present with a predicted difficult airway may also have anatomical or physiological features that make regional anaesthesia difficult or impossible. Advance preparation for regional anaesthetic failure might include: cricothyroid ultrasound and marking;

local anaesthetic application to the airway; flexible nasendoscopy, awake direct or indirect laryngoscopy to assess the view of the larynx.

If primary or secondary (i.e. after failed regional) general anaesthesia is required, the principal decision is whether anaesthesia should be induced before or after securing the airway (Fig. 3). Awake intubation is likely to be the safest option for the woman with an anticipated difficult airway, particularly if difficult or impossible mask ventilation is also anticipated [178]. The recently published DAS guidelines on awake tracheal intubation present a comprehensive description of preparation and performance of awake tracheal intubation [179]. This includes: procedure set up; oxygenation; airway topicalisation; sedation; and a generic awake tracheal intubation technique that can be modified for special circumstances such as the pregnant woman. Awake flexible bronchoscopic intubation is especially indicated if there is restricted mouth opening or a distorted oropharyngeal passage. However, disadvantages include blood or secretions in the airway, or limited time. It is also not a universal skill amongst UK anaesthetists [180]. Videolaryngoscopy is gaining popularity as an awake technique to assess the airway or intubate the trachea [161].

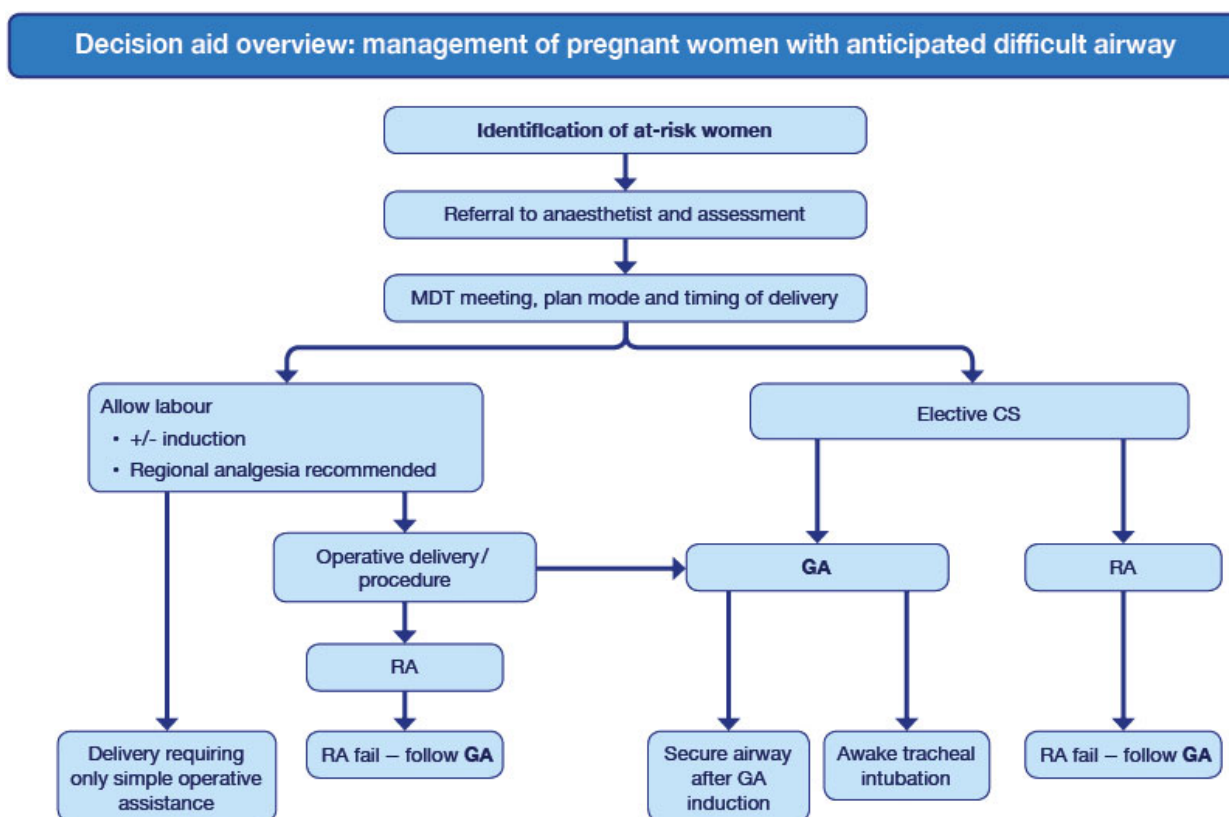


Figure 4 Overview of practice recommendations. GA, general anaesthesia. MDT, multi-disciplinary team. RA, regional anaesthesia; CS, caesarean section; GA, general anaesthesia.

Videolaryngoscopy is likely to be the preferred intubation technique after induction of general anaesthesia in the woman with a predicted difficult airway. This may be more acceptable with the growing availability of high-flow nasal oxygenation in anaesthesia in general, and specifically on the delivery suite. Videolaryngoscopes are now almost universally available in UK obstetric units [181], and have even been suggested as the first-line instrument for routine intubation [165].

Conclusions

The technical aspects of airway management in the pregnant woman are similar to the non-pregnant patient. However, the situational aspects of provision of anaesthesia are very different. Good multidisciplinary teamwork is crucial, and this should commence early during pregnancy or when an airway problem becomes apparent. Figure 4 summarises the overall decision making pathway for a woman with an anticipated difficult airway. Online appendices include a comprehensive compendium of case reports on the management of a number of rare syndromes

and airway conditions (see also Supporting Information, Appendix S7).

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The authors have taken care to confirm the accuracy of information; however, medical knowledge changes rapidly. It is not intended that these management recommendations represent a minimal standard of care during management of the difficult airway, nor should they substitute for good clinical judgement. The application of this information remains the responsibility and professional judgement of the anaesthetist.

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Supporting Information

Additional supporting information may be found online via the journal website.

Appendix S1. Case reports of awake flexible bronchoscopic intubation (AFBI) in pregnant women.

Appendix S2. Case reports of techniques for laryngoscopy and securing the airway while awake in pregnant women.

Appendix S3. Case reports of securing the airway after induction of general anaesthesia in women who had a predicted difficult airway.

Appendix S4. Reports of the use of tracheostomy for management of women with a predicted difficult airway.

Appendix S5. Management of women who had a predicted difficult airway with regional analgesia during labour

Appendix S6. Management of women who had a predicted difficult airway with regional anaesthesia for surgery.

Appendix S7. Case reports of anticipated difficult airway management according to diagnosis. N.B. raised BMI features in some cases that had specific syndromes.